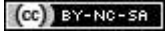


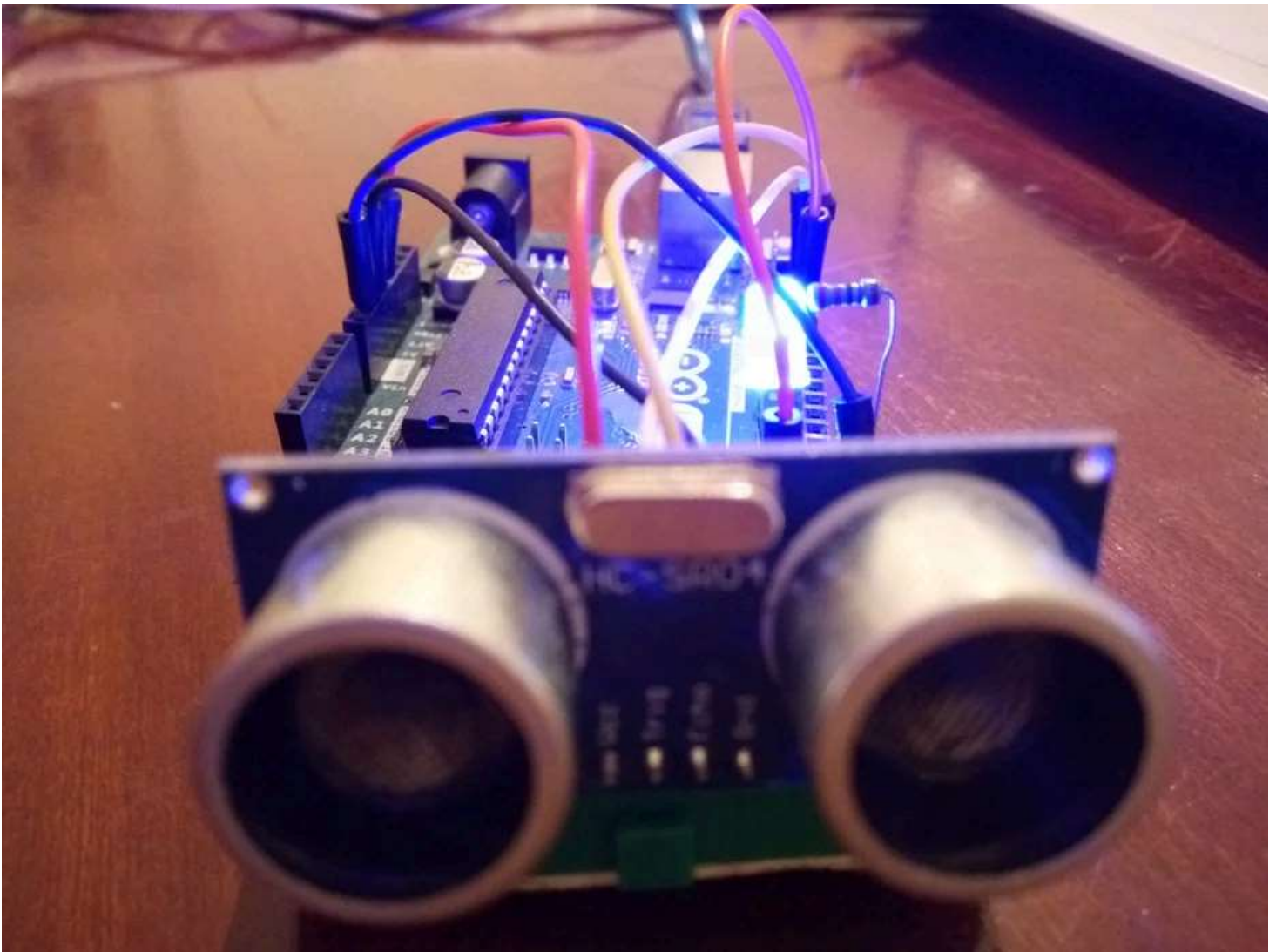
AUTODESK
Instructables

Simple Project With the Ultrasonic Sensor (HC-SR04) +LED - Arduino Tutorial-

By [SaidiDikra](#) in [CircuitsArduino](#)



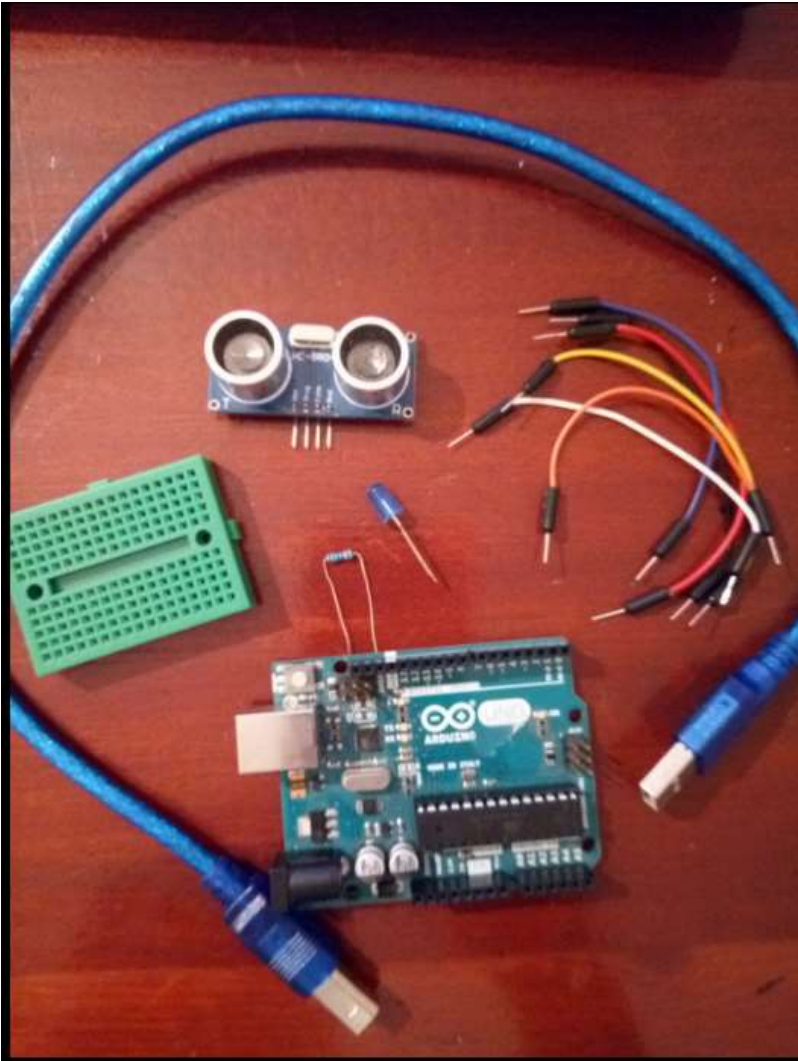
Introduction: Simple Project With the Ultrasonic Sensor (HC-SR04) +LED -Arduino Tutorial-



Ultrasonic ranging module HC - SR04 provides 2cm - 400cm non-contact measurement function, the ranging accuracy can reach to 3mm. The modules includes ultrasonic transmitters, receiver and control circuit.

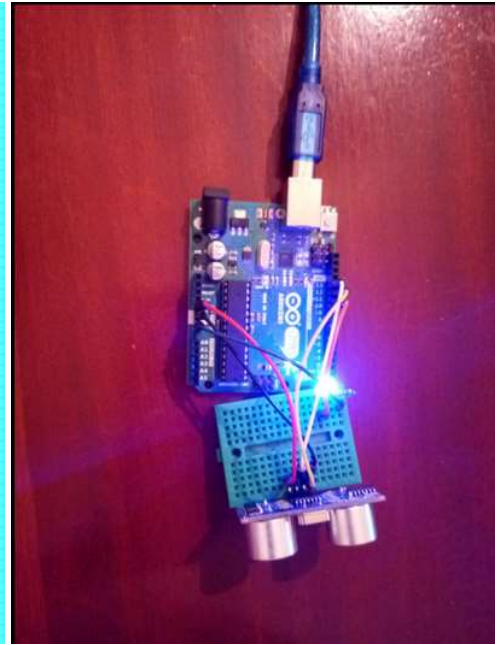
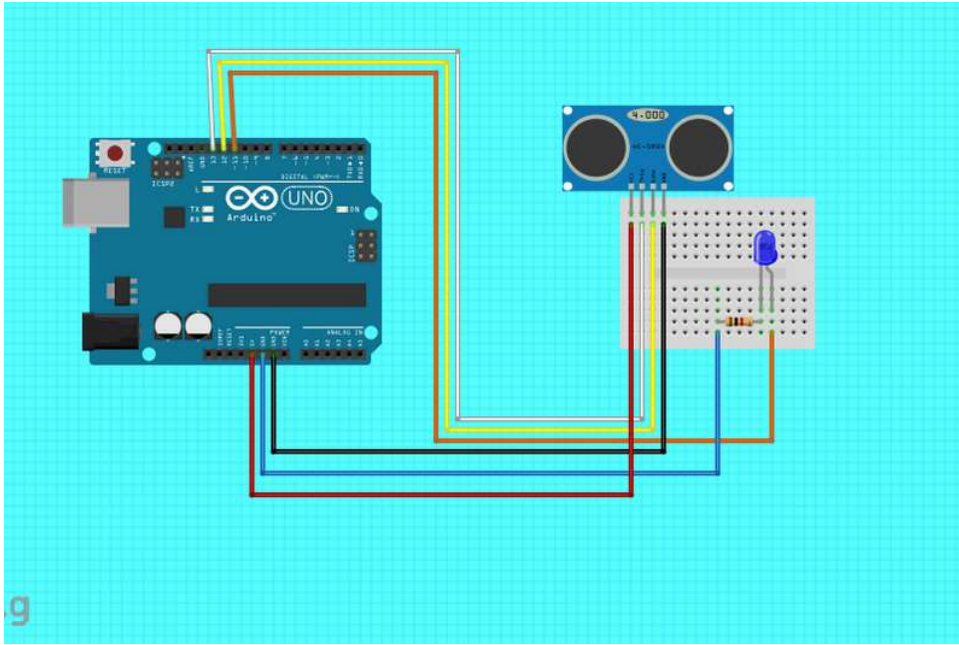
This is a simple example of using the ultrasonic sensor (HC-SR04) in Arduino where we will turn on a led with the variation of distance and print the distance from an object to the serial monitor.

Step 1: All the Necessary Components



1. Arduino Uno
2. Ultrasonic Sensor (HC-SR04).
3. Mini-BreadBoard
4. 1 kohm Resistor.
5. Jumpers.
6. Blue LED.

Step 2: Connect the Components



The connection of the components is very easy it is only necessary to follow the following pictures.

Step 3: Write Your Code

```
#define trigPin 13

#define echoPin 12

#define led 11

void setup()

{ Serial.begin (9600);

pinMode(trigPin, OUTPUT);

pinMode(echoPin, INPUT);

pinMode(led, OUTPUT);

}

void loop()

{ long duration, distance;

digitalWrite(trigPin, LOW);

delayMicroseconds(2);

digitalWrite(trigPin, HIGH);

delayMicroseconds(10);

digitalWrite(trigPin, LOW);

duration = pulseIn(echoPin, HIGH);

distance = (duration/2) / 29.1;

if (distance < 10)

{ digitalWrite(led,HIGH);

}

else {

digitalWrite(led,LOW);

}

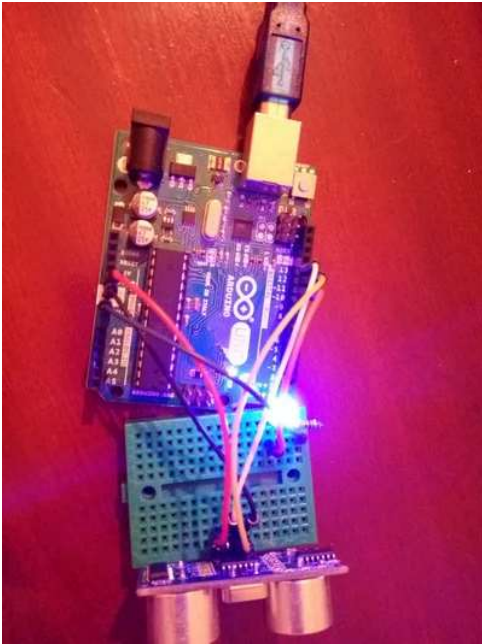
Serial.print(distance);

Serial.println(" cm");

delay(500);
```

```
}
```

Step 4: Upload and Run .



Pas de fin de ligne

You can change the value of distance by replacing 10 in the condition if (distance <=10) by the desired value.